

Declarative Modeling of the U.S. Federal Tax Code with ErgoAI

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Problem Statement

The U.S. federal personal income tax code is complex, highly technical, and filled with exceptions. Existing tax preparation software often uses data-driven LLM or other non-rule based algorithms to assist in tax filing, which can potentially lead to the production of results without clear justification or hallucinations. Individuals and auditors often struggle to understand *why* a particular deduction, credit, or liability applies.

This research project seeks to address this problem using ErgoAI, a rule-based reasoning engine, to encode a significant portion of the 1040 individual income tax process. The system will not only compute taxable income and tax liability but also provide transparent explanations for each line, demonstrating the power of explainable AI in legal and financial domains.

Background

ErgoAI builds on Rulelog/F-logic, offering defeasible reasoning, non-monotonic inheritance, transaction logic, and explainability. This makes it a uniquely strong candidate for encoding the tax code where structured rules and exceptions are in place.

Anticipated Products and Results

We present an implementation of the U.S. federal personal tax code in ErgoAI, a rule-based reasoning platform. Our system encodes the Form 1040 process, including income aggregation, standard vs. itemized deductions, and selected credits. It computes tax liability and refund/amount due while generating human-readable justifications for each step unlike most prosperity standard softwares. We evaluate the system against official IRS worked examples, demonstrating high accuracy and coverage. Our results show that ErgoAI's defeasible reasoning effectively handles exceptions, while its explanation capabilities provide transparency unmatched by traditional tax-filing software. The project contributes to the broader agenda of computational law and explainable AI.

Initial Steps

- Install and configure ErgoAI Studio and the Ergo Reasoner
- Understand how the environment work and play around with sample example and tutorials
- Encode a menial ruleset on some tax-filing examples such as filing status to standard deduction to taxable income
- Run small test cases to confirm correctness and explanations

Low-Risk Results

- A working implementation of filing status and standard deduction rules
- Correct computation of taxable income for simple cases (salary only for now)
- An explanation trace showing how taxable income was computed

Anticipated Results

- A system covering all aspects of Form 1040 - Individual Income Return
 - Income sources
 - Adjusted Gross Income
 - Standard vs. itemized deductions
 - Tax liability from bracket tables
 - Child Tax Credit and Earned Income Tax Credit
 - Refund/amount due from withholding
- A suite of 50-100 cases derived from IRS examples and edge cases
- Explanation for each line of the 1040

Stretch-Goal Results

- Extend coverage to the more tax forms (Form XXXX, Schedule X, etc), including SALT cap, charitable contributes, and mortgage interest
- Implement Education Credit (AOTC, LLC) with income-based phase-outs
- Add counterfactual explanations (“Why didn’t I qualify for XXX?”)
- Live models that updates itself based on new bracket/deduction or changes to the tax code
- Benchmark ErgoAI against other rule-based engines to determine the effectiveness of the software

Relevant Links

Research Idea/Questions to ask - <https://coco.cyberinfrastructure.org/project/207>

Research on Law and Reasoning - <https://ceur-ws.org/Vol-2645/paper5.pdf>

IRS Tax Filing Protocol -

<https://www.irs.gov/e-file-providers/line-by-line-instructions-free-file-fillable-forms>

1040 Tax & EIC Tables booklet - <https://www.irs.gov/pub/irs-pdf/i1040tt.pdf>

2024 IRS Tax Guide - https://www.irs.gov/pub/irs-pdf/p17.pdf?utm_source=chatgpt.com

2024 Instructions for Schedule 8812 -

https://www.irs.gov/pub/irs-pdf/i1040s8.pdf?utm_source=chatgpt.com

2024 Instructions for Earned Income Credit -

https://www.irs.gov/pub/irs-pdf/p596.pdf?utm_source=chatgpt.com